

Python versus R convergence report

Thermal stress and honeybee wing morphology

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Generated automatically by `python check_convergence.py --write-docs`

What this report is

The R pipeline (`R/analysis_main.Rmd`) produced the results reported in the paper. The Python pipeline (`python/run_all.py`) is an independent reimplementaion written to verify them. This report is generated by rerunning both and comparing every quantity the R pipeline reports against the Python recomputation, value by value.

For each comparison the table gives the number of values compared, the largest absolute difference, the largest relative difference (the absolute difference divided by the R value), and the tolerance the comparison is judged against. A row passes when the largest relative difference stays below its tolerance. Tolerances are set far tighter than any decimal place reported in the paper, because the two languages should be performing identical arithmetic; the two places where that expectation is deliberately relaxed are explained in the notes.

Comparisons

Analysis	Quantity	<i>n</i>	Max abs. diff.	Max rel. diff.	Tolerance	Verdict
PERMANOVA	Df	5	0.000×10^0	0.000×10^0	1×10^{-9}	PASS
PERMANOVA	SumOfSqs	5	4.974×10^{-14}	2.333×10^{-14}	1×10^{-9}	PASS
PERMANOVA	R2	5	8.951×10^{-16}	2.350×10^{-14}	1×10^{-9}	PASS
PERMANOVA	F	3	3.581×10^{-12}	2.269×10^{-14}	1×10^{-9}	PASS
PERMANOVA	Pr(>F)					PASS
Follow-up ANOVA (Type II)	SS	15	1.170×10^{-13}	2.296×10^{-13}	1×10^{-8}	PASS
Follow-up ANOVA (Type II)	DF	15	0.000×10^0	0.000×10^0	1×10^{-8}	PASS
Follow-up ANOVA (Type II)	MS	15	1.170×10^{-13}	2.296×10^{-13}	1×10^{-8}	PASS
Follow-up ANOVA (Type II)	F	15	3.587×10^{-11}	2.237×10^{-13}	1×10^{-8}	PASS
Follow-up ANOVA (Type II)	p-unc	15	1.300×10^{-15}	1.593×10^{-11}	1×10^{-8}	PASS
Follow-up ANOVA (Type II)	np2	15	7.008×10^{-15}	2.160×10^{-13}	1×10^{-8}	PASS
Mixed models: estimates	estimate	20	6.237×10^{-9}	2.375×10^{-7}	1×10^{-5}	PASS
Mixed models: AIC/BIC	AIC	20	1.432×10^{-5}	1.075×10^{-9}	1×10^{-6}	PASS
Mixed models: AIC/BIC	BIC	20	1.432×10^{-5}	1.078×10^{-9}	1×10^{-6}	PASS
Mixed models: SE / z / p	std_err	20	5.767×10^{-6}	7.842×10^{-4}	0.001	PASS
Mixed models: SE / z / p	z	20	2.062×10^{-1}	7.836×10^{-4}	0.001	PASS
Asymmetry tests (120 colony x trait)	mean_diff_RL	120	1.041×10^{-16}	3.061×10^{-14}	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	n_pairs	120	0.000×10^0	0.000×10^0	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	t_stat	120	4.974×10^{-14}	4.532×10^{-15}	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	t_pvalue	120	5.551×10^{-16}	3.491×10^{-14}	1×10^{-8}	PASS

Analysis	Quantity	n	Max abs. diff.	Max rel. diff.	Tolerance	Verdict
Asymmetry tests (120 colony x trait)	wilcox_stat	120	0.000×10^0	0.000×10^0	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	wilcox_pvalue	120	4.996×10^{-16}	1.497×10^{-14}	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	t_pvalue_bonf	120	1.499×10^{-15}	3.066×10^{-14}	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	t_pvalue_holm	120	7.772×10^{-16}	3.573×10^{-14}	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	t_pvalue_bh	120	4.441×10^{-16}	3.441×10^{-14}	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	wilcox_pvalue_bonf	120	8.327×10^{-16}	1.344×10^{-14}	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	wilcox_pvalue_holm	120	7.216×10^{-16}	1.520×10^{-14}	1×10^{-8}	PASS
Asymmetry tests (120 colony x trait)	wilcox_pvalue_bh	120	4.996×10^{-16}	9.155×10^{-13}	1×10^{-8}	PASS
PCA	variance explained (%)	5	0.000×10^0	0.000×10^0	1×10^{-9}	PASS
PCA	loadings	25	0.000×10^0	0.000×10^0	1×10^{-9}	PASS
MANOVA (Type III)	Pillai's trace	3	0.000×10^0	0.000×10^0	1×10^{-9}	PASS
MANOVA (Type III)	Wilks' lambda	3	0.000×10^0	0.000×10^0	1×10^{-9}	PASS
Mardia normality test	skewness & kurtosis	5	0.000×10^0	0.000×10^0	1×10^{-9}	PASS
LDA (leave-one-out)	accuracy (%)	3	0.000×10^0	0.000×10^0	1×10^{-9}	PASS
LDA (leave-one-out)	confusion matrix counts	12	0.000×10^0	0.000×10^0	0	PASS
G x E correlations	Pearson & Spearman	20	4.107×10^{-8}	1.000×10^{-7}	1×10^{-6}	PASS
Multiple-comparison corrections	number of significant tests	4	0.000×10^0	0.000×10^0	0	PASS
Robust PCA (MCD)	variance explained (%)					APPROX

Notes

1. **PERMANOVA** / Df: sums of squares computed algebraically, not by building the 4100x4100 distance matrix
2. **PERMANOVA** / Pr(>F): permutation p-values; both reach the 0.001 floor
3. **Mixed models: estimates** / estimate: lme4 and statsmodels use different REML optimisers
4. **PCA** / variance explained (%): R report prints 2 decimals
5. **PCA** / loadings: R report prints 4 decimals
6. **MANOVA (Type III)** / Pillai's trace: R report prints 5 decimals
7. **LDA (leave-one-out)** / confusion matrix counts: every one of about 2,200 bees classified identically
8. **G x E correlations** / Pearson & Spearman: differences are display rounding only: the R report prints the correlation coefficients to 7 significant digits
9. **Robust PCA (MCD)** / variance explained (%): Python RPC1=54.7%, RPC2=21.1% vs R RPC1=54.6%, RPC2=21.1%. sklearn's MinCovDet and rrcov::CovMcd draw different random

subsets; agreement to about 0.1 percentage points is the expected behaviour, not an error

Summary

37 PASS, 1 APPROX, 0 FAIL.

The two implementations are therefore in complete agreement on every quantity that enters the paper. The single APPROX row is the robust PCA, where `sklearn.covariance.MinCovDet` and `rrcov::CovMcd` both implement the Minimum Covariance Determinant estimator but draw different random subsets internally, so exact numerical agreement is not attainable in principle and the comparison is made qualitatively instead.